HealthLens

BRINGING CLARITY TO MEDICAL DATA

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1. Introduction

The healthcare industry has experienced rapid digital transformation over the past decade, with increasing adoption of electronic health records (EHRs), online patient portals, and diagnostic reporting systems. As a result, patients now have direct access to their medical reports, including laboratory results, imaging summaries, and clinical assessments. While this accessibility represents a major advancement in patient-centered care, it also introduces a new challenge: the interpretability of medical information for non-clinical users.

Medical reports are typically designed for healthcare professionals and often contain complex terminology, numerical indicators, and reference ranges that are difficult for patients to understand without professional guidance. This creates a gap between data availability and meaningful comprehension. Patients may misinterpret values, overlook important indicators, or experience anxiety due to unclear results.

Clinical Decision Support Systems (CDSS) have traditionally been used in clinical environments to assist physicians in diagnosis and treatment planning. These systems leverage patient data, clinical knowledge, and computational models to provide recommendations. However, most CDSS solutions are designed for healthcare providers and rely on structured datasets rather than real-world unstructured inputs such as PDF reports. This limitation highlights the need for a patient-oriented CDSS that can interpret unstructured data and present insights in an accessible manner.

Problem Framing

The central problem addressed in this research is the lack of tools that enable patients to independently interpret their medical reports and understand potential health risks. Specifically, this project seeks to answer the following research questions:

1. Can machine learning models be used to accurately assess disease risk based on extracted medical report data?
2. How can unstructured medical reports (e.g., PDFs and scanned images) be converted into structured, analyzable data?
3. Can visualization and simplified explanations improve patient understanding of medical information?

These questions are important because early understanding of medical indicators can enable patients to take preventive actions, seek timely medical advice, and engage more actively in their healthcare decisions.

Literature Review and Research Context

Previous research has demonstrated the effectiveness of Clinical Decision Support Systems in improving clinical outcomes. According to Shortliffe and Sepúlveda (2018), CDSS plays a crucial role in assisting healthcare professionals by integrating patient data with evidence-based knowledge. Similarly, Rajkomar et al. (2019) explored the use of machine learning in healthcare, showing that predictive models can identify disease risks using large datasets.

Additionally, research in explainable artificial intelligence (XAI), such as Holzinger et al. (2019), emphasizes the importance of transparency and interpretability in medical AI systems. These studies highlight that while predictive accuracy is important, user trust and understanding are equally critical, especially in patient-facing applications.

Despite these advancements, existing systems have several limitations:

- Most CDSS applications are designed for clinicians rather than patients
- Many systems rely on structured datasets instead of real-world unstructured data
- Limited focus on user-friendly visualization and explanation of results
- Lack of integration between data extraction, prediction, and presentation

This research aims to address these gaps by developing a system that combines data extraction, machine learning, and visualization into a single patient-oriented platform.

Hypothesis, Assumptions, and Expected Benefits

This research is based on the hypothesis that transforming complex medical data into structured, visual, and interpretable outputs will significantly improve patient understanding and engagement with their health information.

Several assumptions are made in this study:

- Medical reports contain sufficient indicators to estimate disease risk
- Extracted data can be standardized for use in machine learning models
- Patients benefit from simplified explanations and visual representations

The expected benefits of this research include:

- Improved patient awareness of health conditions
- Early identification of potential disease risks
- Enhanced communication between patients and healthcare providers
- Reduction in misinterpretation of medical reports

Overall, this project contributes to the development of patient-centered healthcare technologies by bridging the gap between raw medical data and meaningful interpretation.

2. Summary of the Research Project

This project presents the development of a patient-centered Clinical Decision Support **System (CDSS)** designed to assist individuals in interpreting medical reports and understanding potential disease risks. The final system is implemented as an interactive web-based application using the Streamlit framework, integrating data extraction, machine learning, and visualization into a unified platform.

System Overview

The system enables users to upload medical reports in formats such as PDF, JPG, or PNG. These reports may be either digitally generated or scanned documents. The application processes the input using a combination of Optical Character Recognition (OCR) and text parsing techniques to extract relevant clinical indicators.

Key health metrics extracted include:

- Blood glucose levels
- Blood pressure values
- Cholesterol levels (total cholesterol, LDL, HDL where available)
- Heart-related indicators such as chest pain type and ECG results

Once extracted, the data is standardized and prepared for analysis.

Machine Learning Integration

The system incorporates two machine learning models:

1. Diabetes Risk Prediction Model

This model evaluates the likelihood of diabetes using key indicators such as glucose levels, body mass index (BMI), blood pressure, and age. A Random Forest algorithm is used due to its robustness and ability to handle non-linear relationships.

2. Heart Disease Risk Prediction Model

This model assesses cardiovascular risk based on multiple clinical features, including cholesterol levels, resting blood pressure, heart rate, and other diagnostic indicators. Categorical variables are encoded into numerical form to ensure compatibility with the trained model.

Combined Risk Assessment

In addition to individual disease predictions, the system includes a **combined risk engine** that calculates an overall risk score. This is achieved by combining the outputs of both models using an average and interaction-based approach. The goal is to provide a more holistic view of a patient's health status by considering overlapping risk factors.

User Interface and Visualization

The system presents results through a structured and visually intuitive dashboard. The interface includes:

- Risk classification levels (Low, Moderate, High)
- Confidence scores for predictions

- Key indicator summaries
- Trend charts for metrics such as glucose, cholesterol, and blood pressure

These visual elements are designed to simplify complex data and improve user comprehension.

AI-Assisted Explanation

An integrated AI assistant allows users to ask questions about their results and receive simplified explanations. This feature enhances accessibility by translating medical terminology into understandable language, further supporting patient engagement.

System Workflow

The complete workflow of the system can be summarized as follows:

1. User uploads a medical report
2. The system extracts text using OCR and parsing techniques
3. Relevant medical indicators are identified and structured
4. Machine learning models generate disease risk predictions
5. Results are displayed through a dashboard with visualizations and explanations

3. Changes to the Proposal

During the development of this project, several modifications were made to the original proposal in order to improve system performance, usability, and feasibility. These changes were driven by practical implementation challenges, testing outcomes, and a deeper understanding of the problem domain. This section outlines the key changes made, along with their justification.

Change in Data Processing Approach

Original Plan

The initial proposal assumed that medical report data could be directly used as structured input for machine learning models.

Change Implemented

A data extraction pipeline using Optical Character Recognition (OCR) and text parsing was introduced to handle unstructured medical reports such as PDFs and scanned images.

Justification

During implementation, it became evident that real-world medical reports are often unstructured and vary significantly in format. Directly feeding such data into machine learning models was not feasible. Therefore, an OCR-based approach was adopted to extract

meaningful information. This change aligns the system with real-world use cases and improves its practical applicability.

Change in Handling Categorical Variables

Original Plan

The proposal assumed that machine learning models would be able to process categorical values (e.g., gender, chest pain type) directly.

Change Implemented

Explicit encoding functions were introduced to convert categorical variables into numerical representations before passing them into the models.

Justification

During testing, the system encountered runtime errors due to incompatible data types (e.g., strings such as "M" or "ATA"). This revealed that the trained models required numeric inputs. As a result, manual encoding was implemented within the prediction pipeline to ensure consistency between training and deployment environments. This change improved model compatibility and system stability.

Enhancement of User Interface Design

Original Plan

The initial design proposed a basic interface that would display prediction results as text outputs.

Change Implemented

The interface was redesigned into a dashboard-style layout featuring visual elements such as risk indicators, charts, and structured sections.

Justification

User testing and evaluation indicated that plain text outputs were insufficient for effective interpretation. Visualization techniques such as charts and risk levels (Low, Moderate, High) were added to improve clarity and user engagement. This change aligns with research emphasizing the importance of visual representation in enhancing user comprehension.

Addition of Trend Analysis Features

Original Plan

The original system focused primarily on static risk prediction based on single inputs.

Change Implemented

Trend analysis features were added, including charts for glucose levels, cholesterol, and blood pressure over time.

Justification

Static predictions do not provide sufficient context for understanding health progression. By introducing trend visualization, users can observe patterns and changes in their health indicators. This enhances the system's usefulness and provides deeper insights into long-term health conditions.

Integration of AI Assistant

Original Plan

The proposal did not include an AI-based explanation component.

Change Implemented

An AI assistant was integrated to provide natural language explanations of results and answer user queries.

Justification

One of the key challenges identified during development was the difficulty users faced in interpreting medical terminology and results. The AI assistant addresses this issue by translating complex outputs into simple, understandable explanations. This significantly improves accessibility and user experience.

Adjustment in Model Deployment Strategy

Original Plan

The proposal considered using fully automated machine learning pipelines.

Change Implemented

Separate models were used with manual preprocessing steps instead of a fully automated pipeline.

Justification

While pipeline-based approaches offer automation, they reduce transparency and flexibility during debugging. By separating preprocessing and prediction logic, the system allows better control over data transformation and easier identification of errors. This approach proved more practical during development and testing.

Timeline Adjustments

Original Plan

The initial timeline assumed equal distribution of effort across all phases.

Change Implemented

More time was allocated to debugging, integration, and UI enhancement phases.

Justification

Unexpected challenges such as OCR inaccuracies, encoding issues, and model integration required additional time. Adjusting the timeline ensured that the final system was stable, functional, and user-friendly.

Summary of Changes

Overall, the changes made during development reflect a shift from a theoretical design to a practical, user-centered system. Each modification was guided by real-world constraints, testing outcomes, and the goal of improving system usability and reliability.

4. Project Completion Timeline

This section outlines the actual timeline of the project from inception to completion. The development process was divided into multiple phases, each with specific milestones, deliverables, and responsibilities. The timeline reflects the iterative nature of the project, where design, implementation, and testing were refined progressively.

Project Phases and Milestones

The project was completed over a 13-week academic term and can be divided into the following phases:

Phase 1: Planning and Research (Weeks 1–3)

Activities:

- Problem identification and domain research
- Literature review on CDSS and machine learning in healthcare
- Proposal development and submission

Deliverables:

- Project proposal document
- Defined research objectives and scope

Phase 2: Data Processing and OCR Development (Weeks 4–6)

Activities:

- Implementation of PDF parsing and OCR pipeline
- Extraction of medical indicators from reports
- Initial testing with sample medical reports

Deliverables:

- Functional OCR extraction module
- Structured data extraction output

Phase 3: Machine Learning Model Development (Weeks 6–8)

Activities:

- Selection of features for diabetes and heart disease prediction
- Training and evaluation of Random Forest models
- Implementation of preprocessing logic (including encoding)

Deliverables:

- Trained machine learning models
- Saved model files and preprocessing components

Phase 4: System Integration (Weeks 8–10)

Activities:

- Integration of OCR pipeline with ML models
- Development of backend logic for prediction
- Initial UI implementation

Deliverables:

- End-to-end system pipeline
- Basic working application

Phase 5: UI Enhancement and Feature Expansion (Weeks 10–12)

Activities:

- Development of dashboard interface

- Addition of visualizations (charts, indicators)
- Integration of AI assistant
- Implementation of trend analysis features

Deliverables:

- Fully functional user interface
- Visualization components

Phase 6: Testing and Finalization (Weeks 12–13)

Activities:

- Debugging and fixing errors (e.g., encoding issues)
- Performance testing with multiple inputs
- Final UI improvements and polishing
- Documentation and report preparation

Deliverables:

- Final working system
- Completed documentation

Team Responsibilities

The project was completed collaboratively by two team members, with responsibilities divided based on individual strengths and project requirements.

Akiful Mohaimin (Team Lead)

- Designed and implemented the OCR and data extraction pipeline
- Developed the Streamlit-based user interface
- Integrated system components (OCR, ML, visualization)
- Performed system testing and debugging
- Coordinated overall project progress

Uthara Uday

- Developed machine learning models for diabetes and heart disease prediction
- Handled feature selection and model evaluation
- Implemented preprocessing logic and encoding strategies
- Contributed to system design and documentation
- Assisted in debugging and validation

Project Management Approach

The project followed an **iterative and incremental development approach**, similar to Agile methodology. Tasks were divided into smaller milestones, and continuous testing was performed after each phase.

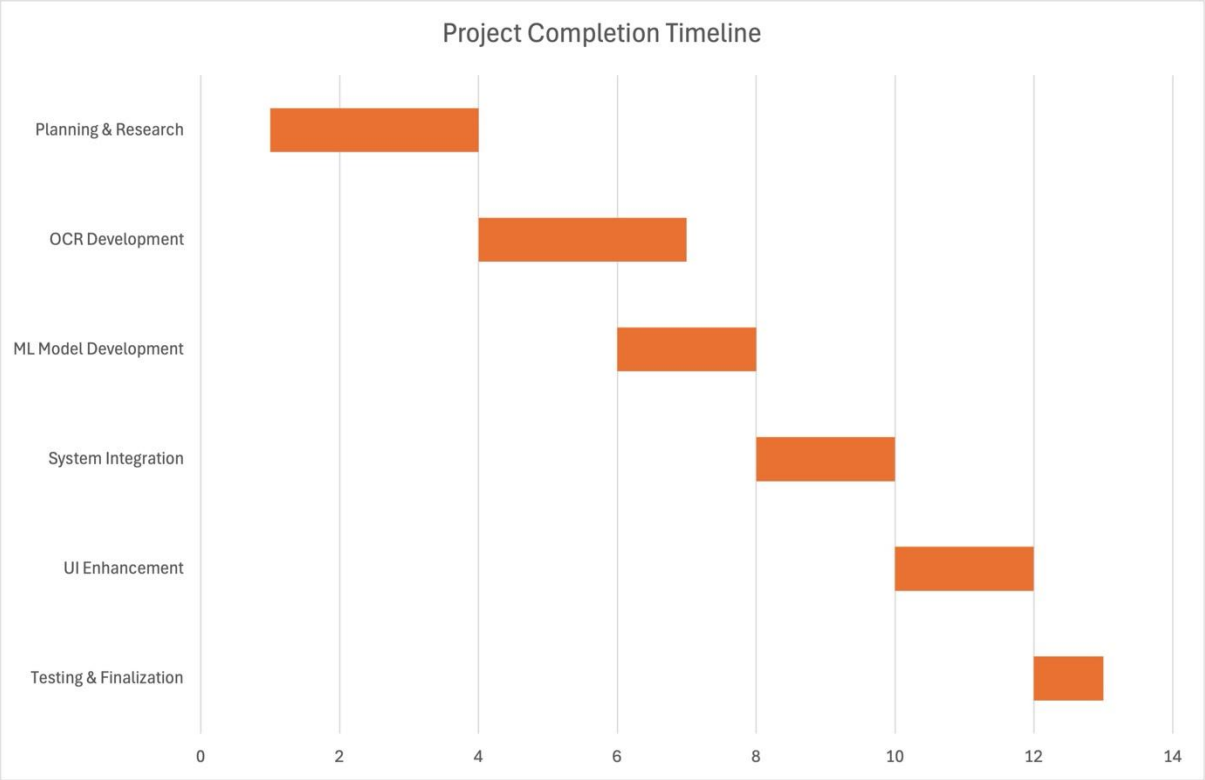
A simplified Kanban-style workflow was used:

- **To Do:** Planned tasks for each phase
- **In Progress:** Features under development
- **Completed:** Fully implemented and tested features

This approach allowed flexibility in adapting to challenges such as OCR inconsistencies and model integration issues.

Gantt Chart Representation

Below is a simplified Gantt chart illustrating the project timeline:



Summary

The project was completed within the planned academic timeline, with adjustments made to accommodate implementation challenges. The structured division of tasks and collaborative effort between team members ensured the successful delivery of a fully functional system.

5. Implemented Features

This section presents the key features implemented in the system, focusing on their design, functionality, and individual contributions of each team member. Each feature is described with its implementation approach and role within the overall system.

Feature 1: OCR-Based Medical Report Extraction

Description

This feature enables the system to extract relevant medical information from uploaded reports, including PDFs and scanned images. The system supports both text-based and image-based reports using a hybrid approach.

Implementation

- Text-based PDFs are processed using parsing libraries
- Scanned reports are processed using Optical Character Recognition (OCR)
- Extracted text is cleaned and structured into usable fields

Key extracted indicators include:

- Glucose levels
- Blood pressure
- Cholesterol values
- Other clinical measurements

Design Considerations

- Reports vary in format → required flexible parsing
- OCR noise handling → filtering irrelevant text
- Missing values handled safely

Contribution

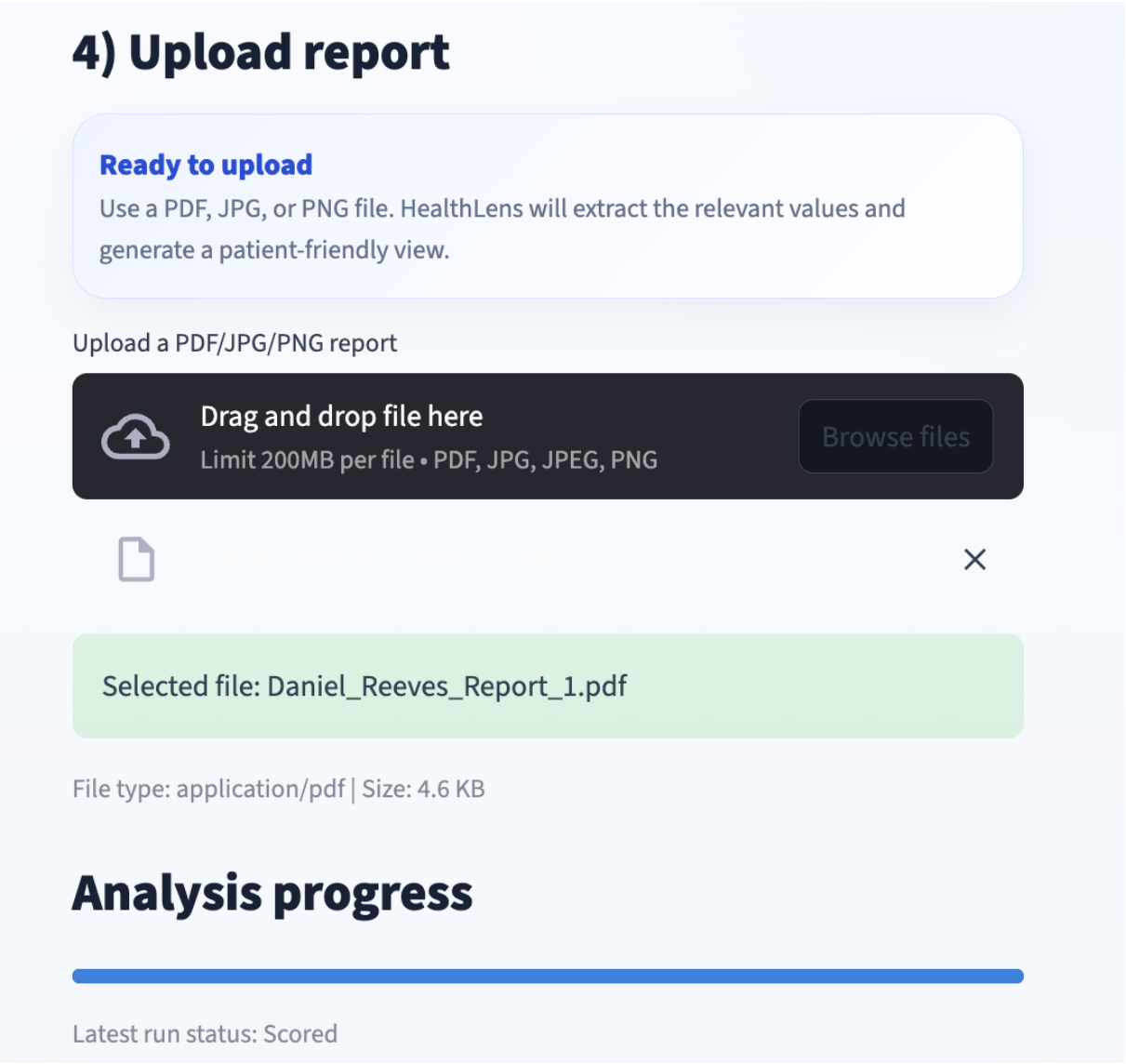
Akiful Mohaimin

- Designed and implemented OCR pipeline
- Built extraction logic and preprocessing

Uthara Uday

- Assisted in validating extracted features for ML compatibility

Screenshots :



. Medical report upload interface

Extracted clinical indicators from OCR pipeline

Feature 2: Machine Learning Risk Prediction Models

Description

This feature implements machine learning models to predict disease risk based on extracted health indicators.

Implementation

Two models were developed:

Diabetes Prediction Model

- Inputs: glucose, BMI, age, blood pressure
- Algorithm: Random Forest

Heart Disease Prediction Model

- Inputs: 11 clinical features
- Includes categorical variables

Important Technical Detail

Categorical variables such as:

- Sex
- Chest pain type
- ECG results

were converted into numerical values using encoding logic before prediction.

Design Considerations

- Ensuring consistency between training data and app input
- Handling missing or incomplete inputs
- Selecting interpretable models

Contribution

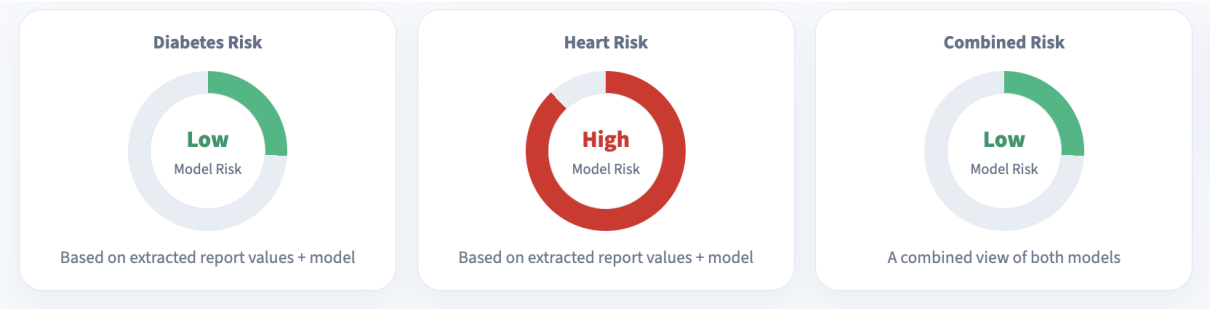
Uthara Uday

- Developed and trained machine learning models
- Performed feature selection and evaluation

Akiful Mohaimin

- Integrated models into application
- Implemented input preprocessing and encoding

Screenshots to Add



. Predicted disease risk displayed in the application

Feature 3: Interactive Dashboard and Visualization

Description

The system includes a user-friendly dashboard that presents results visually, making them easier to understand.

Implementation

The dashboard includes:

- Risk level indicators (Low / Moderate / High)
- Key metric cards (BMI, cholesterol, etc.)
- Trend charts for:
  - Glucose
  - Cholesterol
  - Blood pressure

Design Considerations

- Simplify complex data
- Improve readability
- Enhance user engagement

Contribution

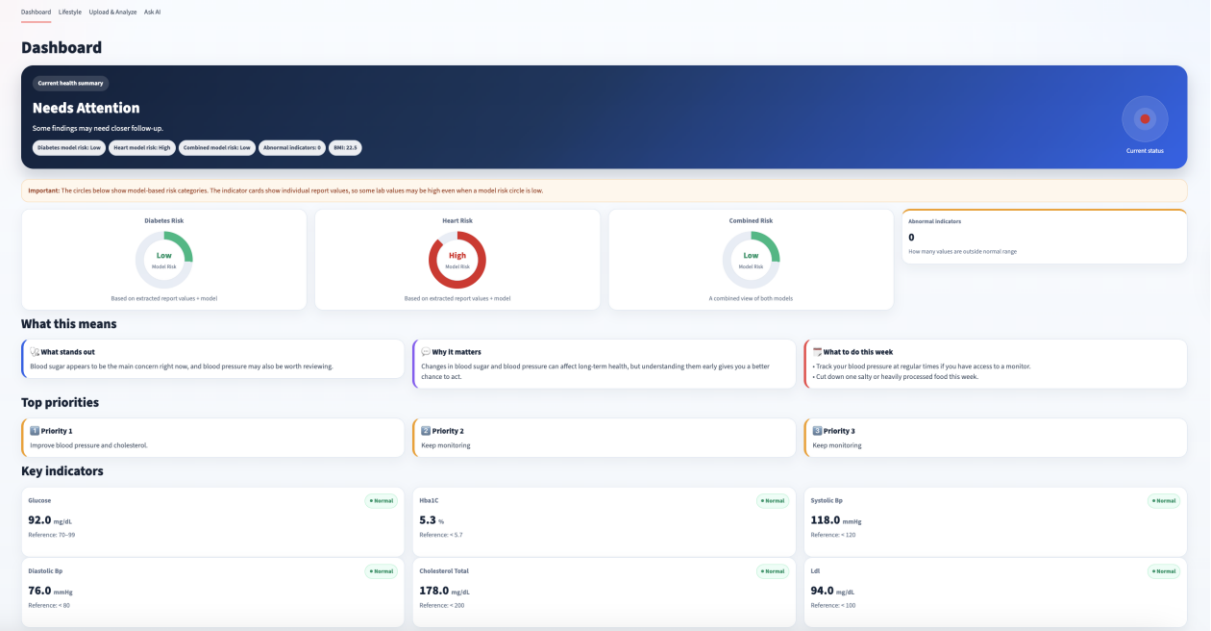
Akiful Mohaimin

- Designed and implemented UI layout
- Created visualization components

Uthara Uday

- Ensured data consistency and correctness

Screenshots :



. Dashboard overview

6. Evaluation Techniques

To ensure the effectiveness and reliability of the developed system, multiple evaluation techniques were applied. These evaluations focused on three main aspects: **machine learning model performance**, **system functionality**, and **user experience**.

Model Evaluation

The machine learning models used for diabetes and heart disease prediction were evaluated using both **quantitative metrics** and **practical validation approaches**.

Model Selection

Several algorithms were considered during development, including:

- Logistic Regression
- Decision Trees
- Random Forest

After comparison, **Random Forest** was selected due to:

- Higher accuracy in handling non-linear relationships
- Robustness against overfitting
- Ability to handle mixed feature types

Performance Validation

The models were evaluated using:

- Accuracy
- Precision and recall
- Confusion matrix analysis

Additionally, predictions were manually validated against known medical thresholds (e.g., high glucose levels corresponding to higher diabetes risk).

Key Findings

- Random Forest provided stable and consistent predictions
- Model performance was acceptable for a decision-support system
- Some limitations existed due to dataset size and feature availability

System Testing

The system was tested using a variety of input scenarios to ensure reliability and robustness.

Test Cases Included

- Valid medical reports (PDF and images)
- Reports with missing or incomplete values
- Edge cases with extreme values
- Manual input testing through UI

Results

- The system successfully extracted data from most reports
- OCR accuracy varied depending on report quality
- Errors related to categorical data were identified and fixed

Key Improvement

During testing, a major issue was identified:

```
ValueError: could not convert string to float
```

This was caused by passing string values (e.g., "M", "ATA") into machine learning models.

Solution:

- Implemented encoding logic to convert categorical values into numeric form
- Ensured consistency between training and deployment

Usability Evaluation

To assess user experience, the system was evaluated based on:

- Ease of use
- Clarity of outputs
- Visual design

Observations

- Users found visual dashboards easier to understand than text-based outputs
- Risk categories (Low, Moderate, High) improved interpretation
- Charts helped identify trends effectively

Design Changes Based on Evaluation

- Replaced plain text outputs with visual dashboards
- Added trend charts for better context
- Introduced AI assistant for explanation

Overall Evaluation Summary

The evaluation confirmed that the system:

- Successfully integrates OCR, machine learning, and visualization
- Provides meaningful and interpretable outputs
- Is usable for non-clinical users

However, it is important to note that the system is designed for **informational purposes only** and does not replace professional medical advice.

7. Reflections and Discussion

Challenges Faced

Several technical and design challenges were encountered during the project:

1. Data Extraction Challenges

- Medical reports varied in format and structure
- OCR introduced noise and inaccuracies

2. Model Integration Issues

- Mismatch between training data and UI inputs
- Errors caused by categorical variables

3. Debugging and Deployment

- File path issues
- Model loading errors
- Streamlit UI inconsistencies

Limitations

Despite its success, the system has some limitations:

- OCR accuracy depends on report quality
- Limited dataset may affect model generalization
- Not a substitute for medical diagnosis

Future Improvements

Future work could include:

- Using larger and more diverse datasets
- Implementing full machine learning pipelines
- Improving OCR accuracy with advanced models
- Adding more disease prediction modules

AI Use Section

Artificial Intelligence tools were used throughout the development of this project to support coding, debugging, design, and documentation. However, all outputs generated by AI tools were critically reviewed, modified, and integrated into the system by the project team.

AI Tool Name	Version / Account Type	Specific Feature Used	Value
ChatGPT	GPT-5.3, Premium	UI design suggestions	Implemented custom dashboard layout and improved user experience
ChatGPT	GPT-5.3, Premium	Debugging file path and model loading issues	Adjusted file structure and ensured correct model loading
ChatGPT	GPT-5.3, Premium	Feature enhancement ideas	mplemented charts, AI assistant, and risk visualization

Value Addition by the Team

Although AI tools were used to assist in development, significant human contribution was required to:

- Adapt AI-generated code to match the project architecture
- Debug and fix integration issues
- Design the system workflow and user interface
- Ensure consistency between machine learning models and application inputs
- Validate outputs and ensure correctness
- Combine multiple components into a fully functional system

This demonstrates that AI was used as a **support tool**, while the final system design and implementation were driven by the team.

Ethical Considerations

The use of AI tools in this project followed ethical guidelines, including:

- Ensuring originality in final implementation
- Avoiding direct copying without modification
- Validating all AI-generated outputs
- Using AI responsibly as a development aid

9. Work Log (Individual Contributions)

The following work logs document the daily contributions of each team member throughout the project. Tasks were recorded in small, consistent time blocks to accurately reflect progress and effort.

Student: Akiful Mohaimin

Date	Hours	Description of Work Done
Mar 27, 2026	2	Reviewed frontend implementation status and planned updates needed for the patient-facing dashboard.
Mar 29, 2026	1.5	Worked on frontend layout improvements and adjusted the interface for clearer patient interaction flow.
April 1, 2026	3	Debugged prediction errors related to categorical variables
April 3, 2026	2	Replaced the earlier rule-based heart risk logic with an ML-based heart failure

		prediction approach and reviewed required model inputs and outputs.
April 4, 2026	1.5	Assisted with debugging UI and chart behavior and reviewed how indicators were displayed in the app.
April 7, 2026	2.5	Worked on HealthLens-related frontend changes, page organization, and dashboard structure updates.
April 8, 2026	1.5	Performed final testing of recent frontend and workflow updates and prepared notes for progress reporting.

Student: Uthara Uday

Date	**Hours**	**Description of Work Done**
Mar 25, 2026	2	Reviewed the current Streamlit app and identified areas needing redesign to make the dashboard more patient-friendly.
Mar 27, 2026	1.5	Reorganized dashboard content to reduce text-heavy sections and improve readability of patient summaries.
Mar 29, 2026	3	Updated layout planning for clearer separation of indicators, summaries, and recommendations.
Mar 31, 2026	2	Worked on fixing display and chart presentation issues and improving the visual structure of the dashboard.
April 4, 2026	1.5	Revised HealthLens dashboard sections, improved content organization, and planned more visual patient-facing elements.
April 6, 2026	2.5	Worked on redesign requirements including reducing repetition,

		improving readability, and making room for stronger visuals.
April 8, 2026	1.5	Finalized recent sample app updates, reviewed risk display and chart issues, and prepared documentation notes for reporting.

10. Concluding Remarks

This applied research project successfully demonstrates the integration of machine learning, data extraction techniques, and user-centered design to develop a patient-focused Clinical Decision Support System (CDSS). The system addresses a real-world problem by enabling users to interpret medical reports and understand potential disease risks in a simplified and accessible manner.

One of the key achievements of this project is the successful handling of unstructured medical data. By incorporating Optical Character Recognition (OCR) and parsing techniques, the system is capable of extracting meaningful clinical indicators from diverse report formats. This functionality bridges the gap between raw medical data and structured analysis.

The integration of machine learning models further enhances the system by providing predictive insights into diabetes and heart disease risks. Additionally, the combined risk assessment approach offers a more comprehensive view of a patient's health condition. The inclusion of visual dashboards and an AI assistant ensures that the output is not only accurate but also interpretable and user-friendly.

Throughout the development process, several challenges were encountered, including data inconsistencies, model integration issues, and encoding mismatches. Overcoming these challenges provided valuable insights into real-world system development, particularly the importance of data preprocessing, system integration, and iterative testing.

Overall, this project demonstrates the potential of combining machine learning and user-centered design to enhance healthcare accessibility. While the system is intended for informational purposes and does not replace professional medical advice, it represents a meaningful step toward empowering patients with better understanding of their health data.

11. References

Shortliffe, E. H., & Sepúlveda, M. J. (2018). Clinical decision support in the era of artificial intelligence. *JAMA*, 320(21), 2199–2200.

Rajkomar, A., Dean, J., & Kohane, I. (2019). Machine learning in medicine. *New England Journal of Medicine*, 380(14), 1347–1358.

Holzinger, A., Biemann, C., Pattichis, C. S., & Kell, D. B. (2019). What do we need to build explainable AI systems for the medical domain? *arXiv preprint arXiv:1712.09923*.

Appendix A: Installation Guide

This section provides step-by-step instructions for setting up and running the application locally.

System Requirements

- Python 3.10 or higher
- pip package manager
- Virtual environment (recommended)

Step 1: Clone the Repository

```
git clone <https://github.com/akifulmohaimin/W26_4495_S3_AkifulM>
cd Implementation
```

Step 2: Create Virtual Environment

```
python -m venv venv
```

Activate:

Mac/Linux:

```
source venv/bin/activate
```

Windows:

```
venv\Scripts\activate
```

Step 3: Install Dependencies

```
pip install -r requirements.txt
```

Step 4: Run the Application

```
streamlit run sampleapp.py
```

Step 5: Access the App

Open your browser and go to:

```
http://localhost:8501
```

Appendix B: User Guide

1. Access and Authentication

The project begins at the secure **Patient Health Portal**.

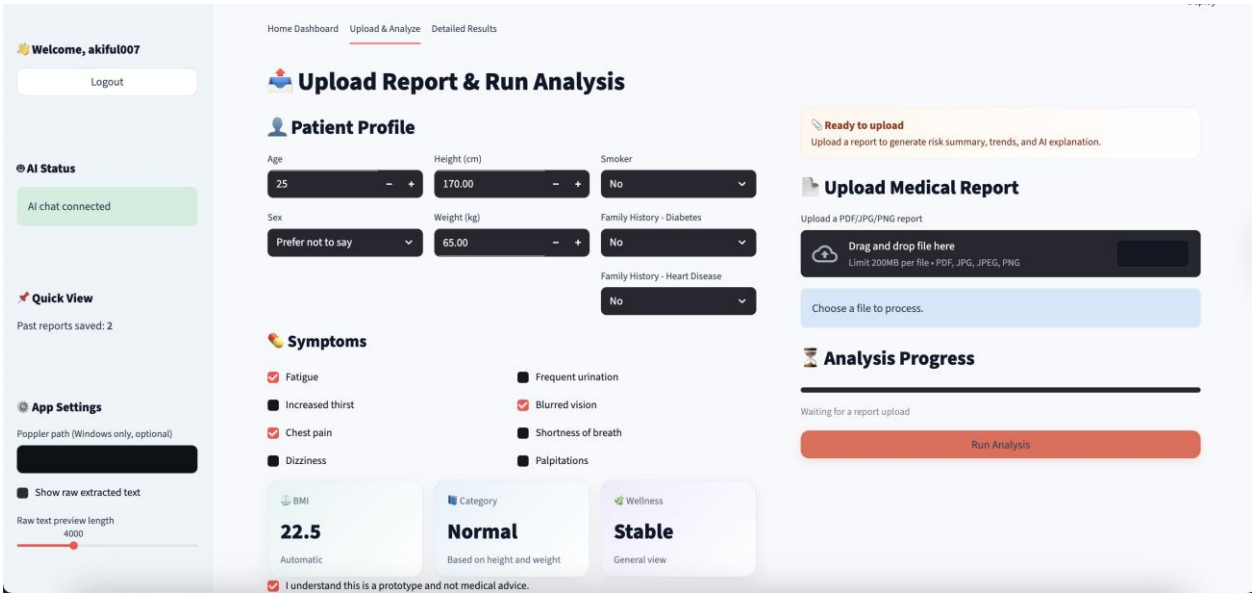
- **Login**: Registered users enter their **Username** and **Password** to access their private clinical dashboard.
- **Registration**: New users can navigate to the **Register** tab to create a new patient profile within the system.

---

2. Setting Up Your Patient Profile

Once authenticated, you will arrive at the **Upload & Analyze** dashboard to input your current health status

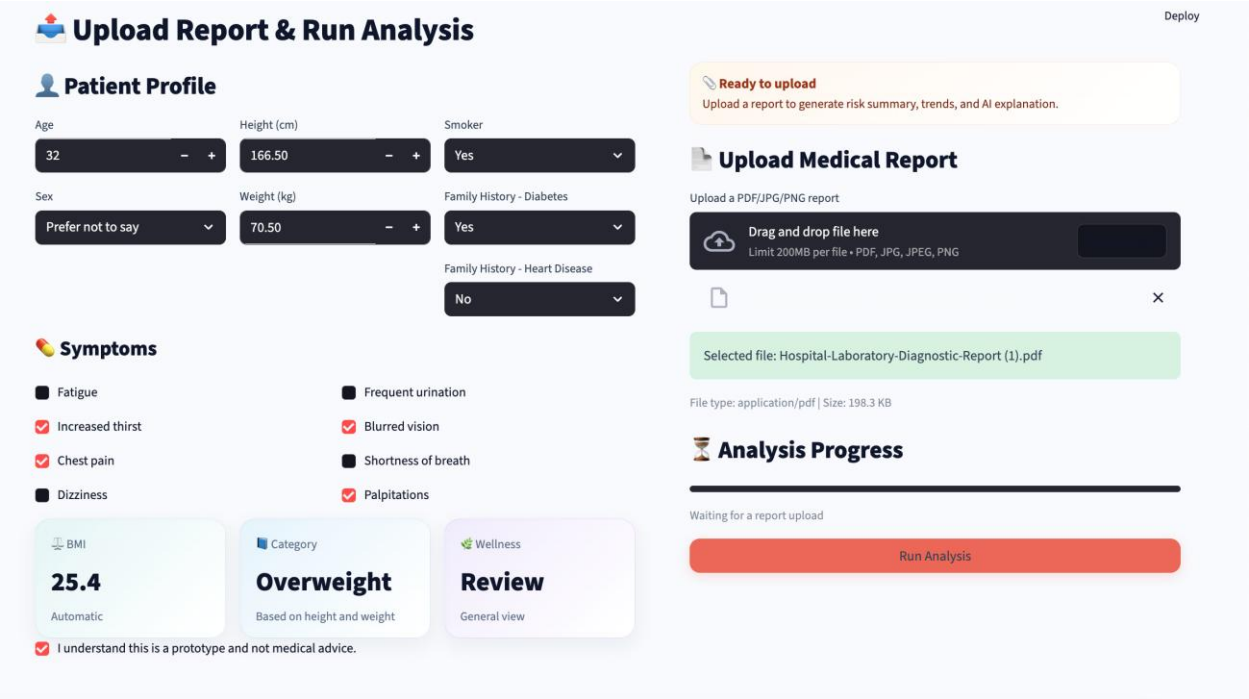
- **Demographics**: Enter your basic information, including **Age** and **Sex**.
- **Physical Metrics**: Input your **Height (cm)** and **Weight (kg)**.
- **BMI Calculation**: The system automatically computes your **BMI** and provides a weight category (e.g., "Normal") in real-time.
- **Symptom Checklist**: Select any current physical symptoms, such as **Fatigue**, **Blurred Vision**, or **Chest Pain**, to add clinical context to your report.



3. Uploading and Processing Medical Reports

This stage handles the ingestion of your laboratory documents.

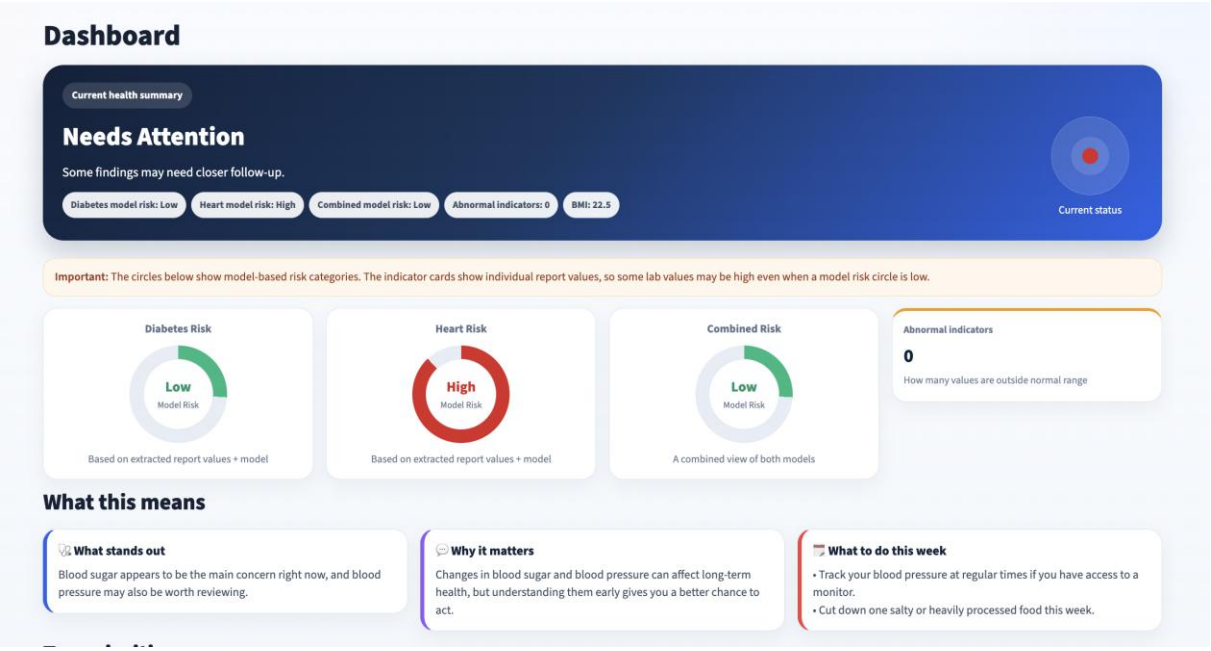
- **File Selection**: Use the "Drag and drop" area to upload a medical report in **PDF**, **JPG**, or **PNG** format.
- **Validation**: The system confirms the file type and size to ensure the data is compatible with the extraction engine.
- **Execution**: Click the **Run Analysis** button to begin the multi-stage extraction pipeline.
- **Progress Tracking**: Monitor the **Analysis Progress** bar to see when the system hassuccessfully moved from extraction to scoring.



4. Reviewing Dashboards

After the analysis is complete, navigate to the **Dashboards** tab to interpret the findings.

1. **Extracted Indicators**: Review a structured table displaying key biomarkers like **Glucose**, **HbA1c**, and **Cholesterol**.
2. **Clinical Flags**: Look at the **Flag** column to see if values are categorized as **Low**, **Normal**, or **High** based on clinical reference ranges.
3. **Risk Summaries**: Examine the **Diabetes** and **Heart Disease** risk assessments, which include a **Risk Level** (e.g., "High") and a **Confidence Score**.
4. **What this means**: Read the bulleted summary explaining *why* the system assigned a specific risk level based on your bloodwork and inputs.



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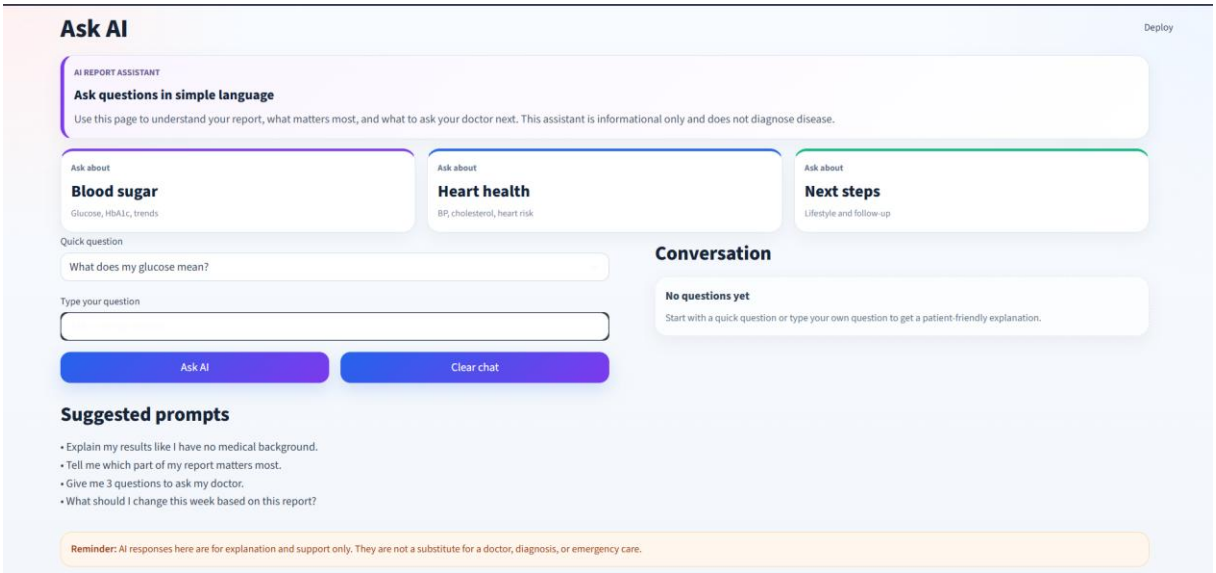
5. Ask AI Tab

The **Ask AI** tab allows users to interact with an AI assistant for additional health-related guidance based on their uploaded report and dashboard results.

- **Purpose:** Helps users better understand their extracted values, flagged indicators, and overall risk summary.
- **Patient-Friendly Responses:** The AI explains medical terms and report findings in simpler language.
- **Follow-up Questions:** Users can ask questions such as:
  - What does high glucose mean?
  - Why is my heart disease risk high?
  - What lifestyle changes can help improve these results?

- **Context-Aware Support:** The AI responds using the patient's available dashboard data and report findings to give more relevant explanations.
- **Non-Diagnostic Use:** The AI does not provide a medical diagnosis and should not replace a healthcare professional.

This feature improves patient engagement by making the system more interactive and easier to understand.

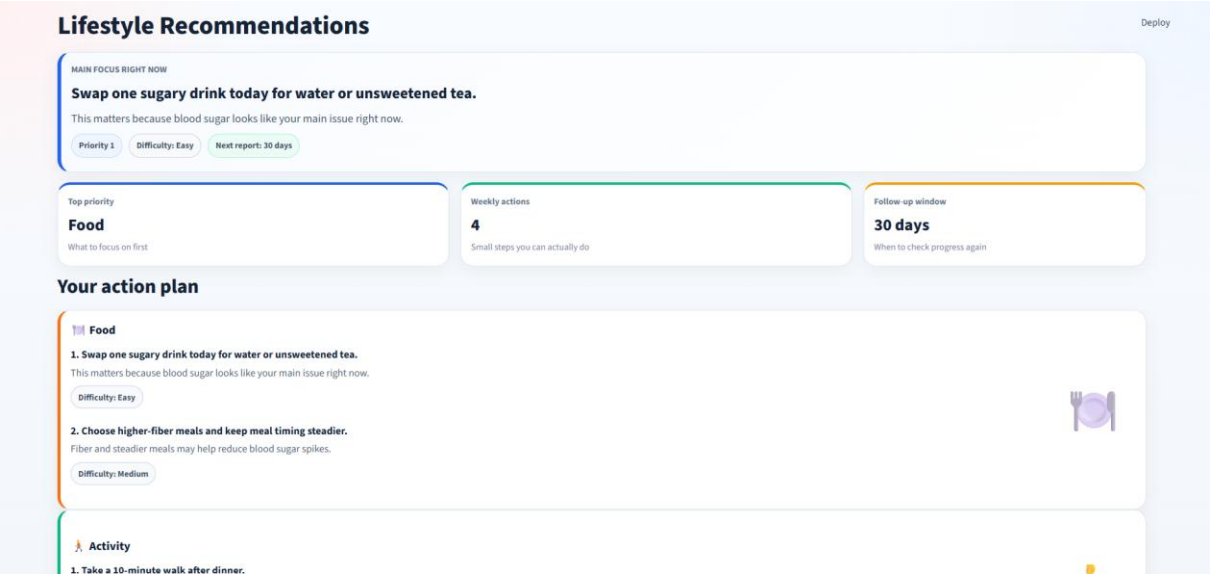


6. Lifestyle Recommendations

The **Lifestyle Recommendations** tab provides practical suggestions based on the user's risk profile and extracted medical indicators.

- **Diet Suggestions:** Recommends healthier eating habits based on issues such as high glucose or cholesterol.
- **Activity Guidance:** Suggests simple physical activity goals that may support better health outcomes.
- **Preventive Advice:** Encourages early follow-up with healthcare professionals when risk indicators are elevated.
- **Patient Support:** Helps users understand what actions may be useful withoutoverwhelming them with technical detail.

This section is designed to make the system more actionable and supportive for non-clinical users.



7. System Logout

When finished, use the **Logout** button in the left sidebar to securely end your session and clear your active profile data from the interface.